WASP-52b

R-0315

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_221206` · created 2026-07-18T00:10:51+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459919.6573 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.216 +/- 0.029
Transit depth (Rp/Rs)^2	4.68 +/- 1.28 [%]
Orbital Inclination (inc)	83.52 +/- 2.07
Airmass coefficient 1 (a1)	1.022 +/- 0.012
Airmass coefficient 2 (a2)	-0.018 +/- 0.067
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	None
Optimal Aperture	4.32
Optimal Annulus	18.46
Transit Duration (day)	0.046 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.216 ± 0.029 vs 0.1646 (deviation 1.77σ)
Duration fit vs published	0.046 d vs 0.0758 d (deviation 1.75σ)
Mid-time O–C	-11.8 min
Depth SNR	7.4
Residual scatter	1.16 %

Light curve

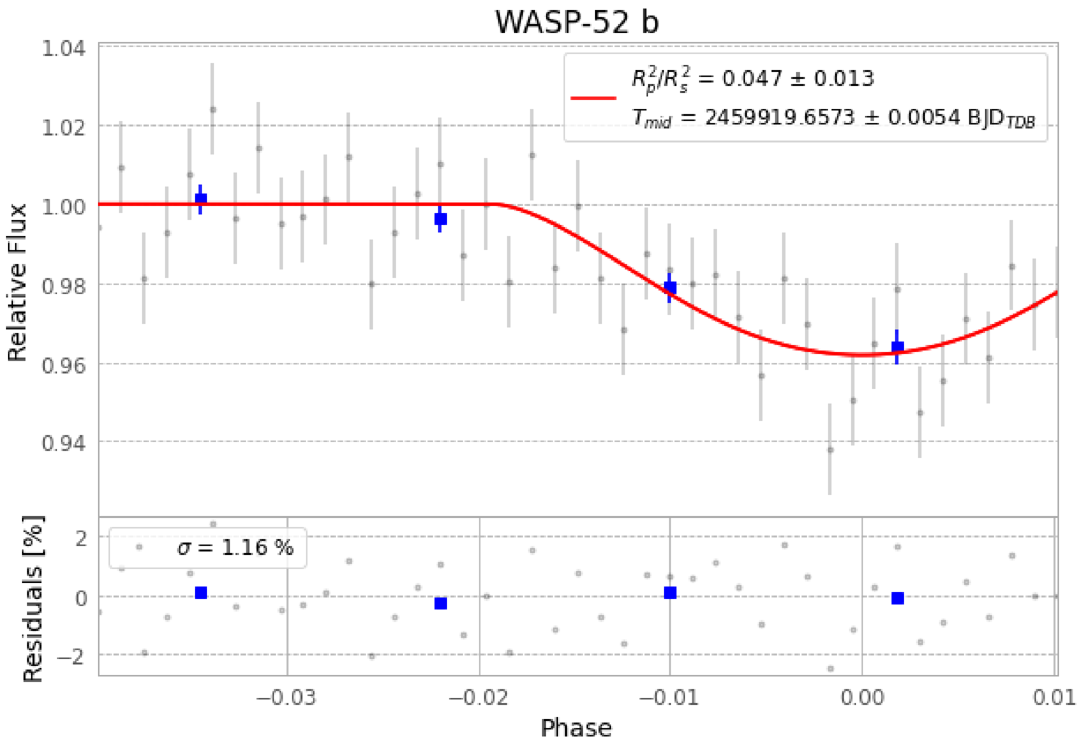

Figure 1 — FinalLightCurve_WASP-52 b_2022-12-05.png (SHA-256 7b64bcbc01401105…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 222], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 545bb65818460e63…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c`

Data provenance

46 FITS frames (30 MB), `WASP-52221206020315.FITS` → `WASP-52221206041815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221206.log (24c2ad40e641…), FinalLightCurve_WASP-52 b_2022-12-05.png (7b64bcbc0140…), FinalLightCurve_WASP-52 b_2022-12-05.csv (0d433d006f68…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)